

1: Functional & Non-Functional Requirements

Functional:

- Upload Videos (Content Creators/Backend)
- Search & Discover videos
- Stream videos smoothly

Non-Functional:

- High Availability (99.999% uptime)
- Extremely Low Latency (Zero buffering)
- High Scalability (Handles millions concurrently)
- High Reliability (No data loss for uploads)

2: Back of the Envelope Estimation

- **Total Users:** ~250 Million
- **Daily Active Users (DAU):** ~100 Million
- **Watch Time:** ~2 hours/day per user
- **Video Size:** ~3 GB per hour (HD)

Bandwidth Estimate:

100M users * 2 hrs * 3GB = **600 Petabytes / day**

Storage Estimate:

Thousands of titles in multiple codecs & resolutions = **Petabytes of raw & processed storage.**

3: Components Used

- **Clients:** Web, Mobile, Smart TV
- **CDN:** Netflix Open Connect (OCA)
- **Load Balancer:** AWS ELB / HAProxy
- **API Gateway:** Zuul API Gateway
- **Microservices:** Auth, Catalog, Search
- **Databases:** Cassandra, ElasticSearch
- **Cache:** Redis / EVCache
- **Storage:** AWS S3 (Raw & Encoded)
- **Event Bus / Analytics:** Apache Kafka & Hadoop

4: Flow and Connections

- 1. Control Plane:** Client requests (login, search, thumbnails) hit the Load Balancer -> API Gateway. Gateway routes to specific microservices.
- 2. Data/Streaming Plane:** When a user hits "Play", the client gets a URL directly to the nearest **Netflix Open Connect (CDN)** server to stream video, bypassing the backend.
- 3. Ingestion Plane:** New content is uploaded to S3 -> Picked up by Transcoders (AWS EC2) to convert into various bitrates/resolutions -> Pushed to Processed S3 -> Distributed globally to CDN.
- 4. Analytics:** All user events (pause, play, search) are streamed to Kafka for Big Data ML processing to generate personalized recommendations.







